

Haar Measure

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Abstract

This is a short expository article on Haar measure on locally compact Hausdorff groups, as a generalization of the Lebesgue measure on $\mathbb{R}$. We also briefly sketch some introduction to its importance in harmonic analysis and probability theory. Lastly, we compute the Haar measure for some important groups.

1 Introduction

The concept of measure theory was introduced by Henri Lebesgue when Vitali proved that there is no set function $\mu : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$ that is translation invariant, countably additive over disjoint sets and assigns every interval its length.

The Lebesgue measure on $\mathbb{R}$ satisfies all the above properties, but is defined on a smaller (yet sufficiently large) collection of measurable subsets of $\mathbb{R}$ that forms a σ -algebra. Following this, we have the general definition of a (positive) measure.

Definition 1.1. A **measure space** is a collection $(X, \mathcal{F}, \mu)$, where X is any set, $\mathcal{F}$ is a σ -algebra on X and $\mu : \mathcal{F} \rightarrow [0, \infty]$ is countably additive over disjoint sets.

Interestingly, an abstract measure does not have all the structures of the Lebesgue measure, even when there is more structure on X than just a set. For instance, the Lebesgue measure preserves the additive group structure and the topological structure of $\mathbb{R}$. Before looking at measures which preserve these structures, let us look at spaces which have these two structures simultaneously.

Definition 1.2. A **topological group** is a group G with a topology such that the following maps are continuous:

$$\begin{aligned} G \times G &\rightarrow G & (g, h) &\mapsto gh \\ G &\rightarrow G & g &\mapsto g^{-1} \end{aligned}$$

Examples of topological groups are $\mathbb{R}^n$, $GL(n, \mathbb{R})$, any group under discrete topology, $\mathbb{Q}_p$ -the field of p-adic numbers.

First we look at measures which are compatible with the topological structure of a set.

Definition 1.3. Let X be any topological space. The **Borel σ -algebra** on X , $\mathcal{B}(X)$, is the smallest σ -algebra containing all open sets.

A measure μ on $\mathcal{B}(X)$ is called **outer regular** if

$$\mu(E) = \inf\{\mu(O) \mid E \subset O \text{ } O \text{ is open}\} \quad \forall E \in \mathcal{B}(X)$$

A measure μ on $\mathcal{B}(X)$ is called **inner regular** if

$$\mu(O) = \sup\{\mu(K) \mid K \subset O \text{ } K \text{ is compact}\} \quad \forall O \text{ open}$$

A measure μ on $\mathcal{B}(X)$ is called a **Radon measure** if μ is inner regular, outer regular, and measure of any compact set is finite.

The above properties are satisfied by the Lebesgue measure on $\mathbb{R}$. Next, we look for measures that are compatible with the group operations.

Definition 1.4. Let G be a topological group. A measure μ on $\mathcal{B}(G)$ is called **left-invariant** if

$$\mu(sE) = \mu(E) \quad \forall E \in \mathcal{B}(G), s \in G$$

Similarly, define right-invariant.

Definition 1.5. Let G be a topological group. A measure μ on $\mathcal{B}(G)$ is called a left\right **Haar** measure if it is Radon and left\right translation invariant.

We will now state an important theorem regarding Haar measures.

Theorem 1.1. If G is a locally compact Hausdorff (LCH) topological group, then there exists a left Haar measure that is unique up to positive scalar multiples.

Proof. Theorem 1.3.4 in [1] □

Remarks:

1. For readers who are not familiar with topological spaces, any metric space is Hausdorff. In fact, Hausdorff condition is not necessary for the theorem but locally compact condition is necessary.
2. Even in the case of $\mathbb{R}$ it shows that the measure satisfying the original three conditions must be the usual Lebesgue measure. (The scalar multiple in the theorem is normalized such that measure of an interval is its length).

2 Properties of Haar measure

In this section, G denotes an LCH group, and any measure would be assumed to be a non-zero Radon measure.

Proposition 2.1. μ is a left Haar measure iff $\tilde{\mu}(E) := \mu(E^{-1})$ is a right Haar measure.

Proof. Suppose μ is a left Haar measure. For any Borel set $E \in \mathcal{B}(G)$ and any $x \in G$, we evaluate the right translate of E :

$$\tilde{\mu}(Ex) = \mu((Ex)^{-1}) = \mu(x^{-1}E^{-1})$$

Because μ is left translation invariant, $\mu(x^{-1}E^{-1}) = \mu(E^{-1})$. By definition, $\mu(E^{-1}) = \tilde{\mu}(E)$. Thus, $\tilde{\mu}$ is right-invariant. Furthermore, since the inversion map $g \mapsto g^{-1}$ is a homeomorphism, $\tilde{\mu}$ naturally inherits the Radon property from μ . The converse follows by a symmetric argument. $\square$

Proposition 2.2. μ is a left Haar measure iff

$$\int_G f(x^{-1}g)d\mu(g) = \int_G f(g)d\mu(g) \quad \forall x \in G, f \in L^1(G)$$

Proof. ($\Rightarrow$) Suppose μ is a left Haar measure. Let E be a Borel set with finite measure and χ_E be its indicator function. Notice that $\chi_E(x^{-1}g) = 1$ if and only if $x^{-1}g \in E$, which means $g \in xE$. Integrating this yields:

$$\int_G \chi_E(x^{-1}g)d\mu(g) = \int_G \chi_{xE}(g)d\mu(g) = \mu(xE)$$

Since μ is left-invariant, $\mu(xE) = \mu(E) = \int_G \chi_E(g)d\mu(g)$. By linearity, this equality extends to all simple functions in $L^1(G)$. Since simple functions are dense in $L^1(G)$, the equality holds for all $f \in L^1(G)$ via standard limiting arguments.

($\Leftarrow$) Suppose the integral equality holds for all $f \in L^1(G)$. Let $E \in \mathcal{B}(G)$ be a set of finite measure. Then its indicator function $\chi_E \in L^1(G)$, and we can apply the hypothesis:

$$\mu(xE) = \int_G \chi_{xE}(g)d\mu(g) = \int_G \chi_E(x^{-1}g)d\mu(g) = \int_G \chi_E(g)d\mu(g) = \mu(E)$$

Thus, μ is left translation invariant for all sets of finite measure. For sets of infinite measure, left invariance follows by taking supremums over compact subsets (which have finite measure) due to the inner regularity of the Radon measure μ . Therefore, μ is a left Haar measure. $\square$

Proposition 2.3. For any left Haar measure μ , $\mu(G) < \infty$ iff G is compact.

Thus given any compact Hausdorff group, can normalize the Haar measure such that the measure of the whole group is 1. This is important in the context of probability theory, because this answers which sets have a ‘meaningful’ notion of a uniform probability measure. For instance, we can choose uniformly from $SO_n(\mathbb{R})$, S^n , $\mathbb{T}^n$, but we cannot choose uniformly from $SL_n(\mathbb{R})$ or $Pos_n(\mathbb{R})$.

We state two lemmas to prove the above proposition.

Lemma 2.4. If μ is a Haar measure $\mu(O) > 0$ for any non-empty open subset O of G .

Proof. If $\mu(K) = 0$ for all compact subsets K , then by inner and outer regularity, $\mu(E) = 0$ for every $E \in \mathcal{B}(G)$, which contradicts that $\mu \neq 0$. Choose K_0 compact such that $\mu(K_0) > 0$, and U open non-empty set. Then

$$K \subset G = \bigcup_{s \in G} sU$$

Since K is compact $\exists s_1, \dots, s_n$ such that

$$K \subset \bigcup_{i=1}^n s_i U$$

Thus, $0 < \mu(K) \leq n\mu(U)$. □

Lemma 2.5. Every open neighborhood U of e contains a neighborhood V of e such that $V^2 \subset U$.

Proof. Define a map,

$$\phi : U \times U \rightarrow G \quad (u, v) \mapsto uv$$

$\phi^{-1}(U)$ is open and contains (e, e) . Thus, $(e, e) \in V_1 \times V_2$ for some open sets V_1, V_2 . Let $V = V_1 \cap V_2$, then it is easy to check $V^2 \subset U$. □

Proof of proposition (2.3)

Proof. If G is compact then, $\mu(G) < \infty$ by property of Radon measure.

For converse, assume $\mu(G)$ is not compact. Since G is locally compact, $\exists$ open $U \ni e$ such that $\bar{U}$ is compact. By previous lemma, choose V such that $V^2 \subset U$. Since finite union of compact sets is compact, no finite translates of K cover G . Choose a sequence $\{s_j\}$ such that

$$s_n \notin \bigcup_{j < n} s_j K$$

We claim that $\{s_j(V \cap V^{-1})\}$ is a disjoint collection of open sets, from which the conclusion follows.

For contradiction assume $i < j$ and $u, v \in V \cap V^{-1}$ such that $s_i u = s_j v$. Then $s_j = s_i u v^{-1} \in s_i K$, which contradicts the choice of $\{s_j\}$. □

Example 2.1. Let $X \sim Unif(O_n(\mathbb{R}))$, then

1. What is $\mathbb{P}(X \in SO_n(\mathbb{R}))$?
2. What is $\mathbb{E}(\det(X))$?

Example 2.2. Let $X \sim Unif(U_n(\mathbb{C}))$, then

1. Find the distribution of $\det(X)$.
2. What is $\mathbb{E}(\det(X))$?

Solution. Let $M = \text{Diag}(\exp(i\theta), 1, \dots, 1) \in U_n(\mathbb{C})$. Then

$$MX \stackrel{d}{=} X$$

Hence, $e^{i\theta} \det(X) \stackrel{d}{=} \det(X)$. Notice that $\det(X)$ is an S^1 -valued random variable whose distribution is invariant under translation by any element of S^1 . By uniqueness of Haar measure $\det(X) \sim \text{Unif}(S^1)$.

Also taking expectation in the above equality, $e^{i\theta} \mathbb{E}(\det(X)) = \mathbb{E}(\det(X))$. Choose $\theta \neq 0$, then $e^{i\theta} \neq 1$, hence $\mathbb{E}(\det(X)) = 0$ $\square$

Example 2.3. If $(X_1, X_2) \sim G_1 \times G_2$, then show that $X_1 \perp\!\!\!\perp X_2$.

Another important property of a left Haar measure on a compact group is that it is automatically right-invariant, and hence a right Haar measure, however this is not a characteristic property of compact groups unlike previous proposition. To establish the right-invariant property, we define the modular function.

Given a left Haar measure μ on G and an element $x \in G$, define

$$\mu_x^r(E) = \mu(Ex)$$

It is easy to check that μ_x^r is another left Haar measure. Since left Haar measure is unique upto scalar multiples, there exists $\Delta_\mu(x)$ such that

$$\mu_x^r(E) = \Delta_\mu(x) \mu(E)$$

Check that Δ_μ is independent of μ . Thus Δ defines a function $\Delta : G \rightarrow \mathbb{R}^+$, which is called the modular function of G .

Proposition 2.6. Δ is a continuous group homomorphism from G to $\mathbb{R}^+$.

Proof. Lemma 1.4.2 in [1] $\square$

Proposition 2.7. If G is a compact Hausdorff group, any left Haar measure is also a right Haar measure.

Proof. Since Δ is continuous and G is compact, $\Delta(G)$ is a compact subgroup of $\mathbb{R}^+$. But $\mathbb{R}^+$ has only one compact subgroup, $\{1\}$. Therefore, $\Delta(g) = 1 \ \forall \ g \in G$. Hence,

$$\mu(Eg) = \mu_g^r(E) = \Delta(g) \mu(E) = \mu(E)$$

So μ is right-invariant, and hence a right Haar measure. $\square$

3 Introduction to Harmonic Analysis of Abelian groups

The basic idea of Fourier analysis is to approximate functions through nice functions which form an orthogonal basis. There are two preliminary examples of this phenomenon: one is the discrete Fourier series which expresses functions from $\mathbb{R}/\mathbb{Z} \rightarrow \mathbb{C}$ as a series

$$f(x) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{(2\pi i n x)}$$

and the second is expressing functions from $\mathbb{R} \rightarrow \mathbb{C}$ as an integral

$$f(x) = \int_{\mathbb{R}} \hat{f}(z) e^{2\pi i x z} dz$$

Fourier analysis over finite Abelian groups arose in the work of Dirichlet in his famous work on primes in an arithmetic progression. See [2] for details. Now that we have a meaningful measure to integrate over any LCH group, we can ask a similar question in the abstract setting of an LCH group. This is the study of harmonic analysis. We express the idea in a much simpler case of an Abelian group.

Definition 3.1. A (unitary) **character** of G is a continuous group homomorphism $\chi : G \rightarrow S^1 \subset \mathbb{C}$.

The **Dual** of G is defined as

$$\hat{G} = \{\chi : G \rightarrow \mathbb{C} \mid \chi \text{ is a character}\}$$

It can be seen that $\hat{G}$ is an Abelian group under point-wise multiplication, however it is not immediate that it is LCH. This is the theory of Pontryagin duality (which we omit) that defines the topology on $\hat{G}$ that makes it LCH, and hence equips it with a Haar measure $\hat{\mu}$.

Definition 3.2. Given a character χ and $f \in L^1(G)$, define the Fourier transform

$$\hat{f}(\chi) = \int_G f(x) \bar{\chi}(x) dx$$

Proposition 3.3. For appropriate class of functions, we have the Fourier expansion,

$$f(x) = \int_{\hat{G}} \hat{f}(\chi) d\hat{\mu}(\chi)$$

The major theorems in classical Fourier analysis such as Fourier inversion, Plancherel Theorem, Poisson summation formula extends appropriately to this setup. If G is non-abelian but compact there is still a general theory (via irreducible representations) due to Peter-Weyl theorem, however non-abelian and non-compact groups are extremely difficult to analyze in this manner other than some specific cases. See [1] for Abstract harmonic analysis.

4 How to construct the Haar measure?

In this section, we give a method to compute the Haar measure in one specific case.

Proposition 4.1. If G is an LCH group, such that $G \subset_{open} \mathbb{R}^n$ for some n with subspace topology (it need not be a subgroup). Further assume,

$$M_x^l : G \rightarrow G \quad g \mapsto xg$$

is C^1 with non-zero derivative.

If the determinant of the Jacobian matrix is a constant for every x , define $S_l(x) = |\det(DM_x^l)|$. Then $\frac{dx}{S_l(x)}$ is a left Haar measure on G .

Proof. f is absolutely continuous wrt Lebesgue measure hence it's a Radon measure. We only need to verify left-invariance. Observe that

$$D_x M_{g_1 g_2}^l = D_x (M_{g_1}^l (M_{g_2}^l)) = D_{g_2 x} (M_{g_1}^l) D_x (M_{g_2}^l)$$

Thus,

$$S_l(g_1 g_2) = S_l(g_1) S_l(g_2)$$

For any $f \in C_c^+(G)$, and any $g \in G$, we have,

$$\begin{aligned} \int_G f(x) \frac{dx}{S_l(x)} &= \int_G f(M_g^l(x)) \frac{|\det(D_x M_g^l(x))|}{S_l(M_g^l(x))} dx \\ &= \int_G f(gx) \frac{S_l(g)}{S_l(g) S_l(x)} dx \\ &= \int_G f(gx) \frac{dx}{S_l(x)} \end{aligned}$$

Hence proved. □

Remark: The concept of C^1 maps is present in differentiable manifolds, where chain rule and change of variable theorem also hold. Thus the proposition can also be extended to the case G is a sub-manifold of $\mathbb{R}^n$.

References

- [1] Deitmar, Anton, and Siegfried Echterhoff. *Principles of Harmonic Analysis*. Springer, 2nd edition, 2014.
- [2] Stein, Elias M., and Rami Shakarchi. *Fourier Analysis: An Introduction*. Princeton University Press, 2003.